

VR-Forms. Preprint.

Part VIII. Open Questions and the Place of VR-Forms in the Cycle

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VIII.0. Introduction

The concluding part of the preprint sums up VR-Forms and records open directions. The structure parallels Part IX of VR-Sets: technical questions, substantive extensions, philosophical open directions, general conclusion.

Part VIII closes the work. All formal results are gathered in Parts II–IV; applications in Parts V–VI; logical properties in Part VII. Here — the fixing of what remains open and of what is planned as the next work of the cycle (VR-Audit).

VIII.1. Technical open questions

Question 1: formalisation in Lean/Coq/Agda

Machine formalisation of VR-Forms is a natural technical step. The two-register structure fits well with dependent types: the ontological register — usual types with values, the formal register — syntactic objects with an interpreting mapping. Lean 4 appears most suitable: its mathematical library (mathlib) already contains tools for working with conservative extensions, translations, two-sorted logic.

Concrete formalisation tasks: (a) formal verification of Theorem III.1 on conservativity; (b) formalisation of the transit rule (§IV.2); (c) machine verification of examples (§IV.5, Parts V–VI).

Question 2: precise strength of the translation π

The mapping π from §III.2 translates formal terms into defining predicates. On simple formal terms of the form $\ulcorner \{x : \varphi(x)\} \urcorner$ the translation is trivial. For more complex constructions (quantification over formal terms, iteration of formal terms, recursive definitions via formal terms) the translation requires more careful definition.

Open question: what is the precise range of formal constructions for which π gives an invertible translation? Are there formal terms for which π gives an ambiguous result? These questions matter for a strict variant of formalisation in Lean.

Question 3: composition of transits

If there is a transit D_1 yielding the ontological statement φ , and a transit D_2 using φ as a lemma on the way to a statement ψ — is there a single transit for ψ , or is a composition of separate transits needed?

This is a question of practical work with VR-Forms: how naturally do chains of lemmas organise into one large transit, and do problems with conservativity arise under composition?

It is expected that the composition of transits is safe (by transitivity of conservativity), but a strict proof and a survey of practical cases remain open.

Question 4: alternative logics of the formal register

In §VII.1 the choice was made in favour of classical logic in the formal register. Open question: what properties of VR-Forms are preserved, and which change, under replacement of classical logic with:

- (a) intuitionistic logic — substantively: the formal register becomes constructive; paradoxical formal terms receive a different status;
- (b) paraconsistent logic — substantively: contradictions in formal terms may be treated explicitly, without threat of overall contradiction;
- (c) modal logic — substantively: formal terms may differ in «mode» (necessary/possible/counterfactual).

Each of these alternatives is a separate work.

VIII.2. Substantive extensions

Extension 1: VR-Audit

The direct and principal continuation of VR-Forms is the programme of VR-Audit. Content: systematic analysis of classical mathematical theorems with the aim of determining which of them transit directly into VR-Sets, which require operational labour to extract content (problematic transits), and which are essentially non-ontological.

VR-Forms provides the apparatus for VR-Audit. VR-Audit uses this apparatus on concrete theorems. List of first candidates:

- the spectral theorem in its various formulations (discrete and continuous spectrum);
- the Hahn–Banach theorem in general form;
- Baire's category theorem;
- Tikhonov's compactness theorem;
- the Radon–Nikodym theorem on measures;
- theorems from quantum mechanics (Stone, von Neumann, Naimark).

Each such theorem can be analysed: is the uncountable used essentially, or only as convenience?

Extension 2: connection with reverse mathematics

The programme of Reverse Mathematics (Friedman, Simpson) classifies classical theorems by the strength of axioms needed for their proof. The hierarchy of subsystems of second-

order arithmetic (RCA_0 , WKL_0 , ACA_0 , ATR_0 , $\Pi^1_1\text{-CA}_0$) gives a precise measure of the «uncountable strength» of each theorem.

VR-Forms and VR-Audit can connect with this programme through the translation π . Each system of the Reverse Mathematics hierarchy corresponds to a definite class of formal terms in VR-Forms; theorems provable in RCA_0 transit directly into VR-Sets; theorems requiring stronger subsystems yield problematic transits, and the character of problematicity depends on the subsystem.

Establishing the precise correspondence is a separate task, uniting VR-Forms with the already substantial Reverse Mathematics literature.

Extension 3: connection with topoi and category theory

Categorical set theory (Lawvere ETCS, Macnamara SDG, elementary topoi) gives an alternative view of foundations through universal properties. VR-Sets is close to them in spirit (see VR-Sets §IX.2). VR-Forms adds the possibility of speaking about non-operational categorical constructions as formal terms: «topos of all functors», «absolutely complete category», etc.

Open direction: construction of VR-categorical apparatus, where operational categories (countable, with countable structure) are operational, and larger ones are formal.

Extension 4: homotopy type theory (HoTT)

In HoTT equality is not a relation but a type (path). VR-Forms can be reinterpreted in the HoTT spirit: $\equiv$ as a path in the type of operational sets, formal terms as higher types with a special interpreting translation. This is technically complex but substantively close.

Extension 5: operational foundations of physics

In VR-Sets §IX.3 the question of the applicability of operational ontology to quantum mechanics was raised. VR-Forms provides a more complete apparatus for this direction.

Hypothesis: quantum mechanics of operational systems (with discrete spectrum of the Hamiltonian) is fully formulated in VR-Sets; quantum mechanics of continuous systems requires the formal register for its uncountable constructions (classical Hilbert space, spectral measure, projection-valued measure). At the same time the physically observable results — measurable quantities — should transit back into the operational.

Verification of this hypothesis on concrete physical examples is a task leading to a possible future work VR-Physics.

VIII.3. Philosophical open questions

Question 1: status of the formal term

In Part I §I.4 it was fixed: a formal term is simply a formula, without ontological status. This position is consistent but philosophically rigid. Open question: can an intermediate status of formal terms be found — for example, «existing in the sense of use», «existing as linguistic objects», «existing in Popper's third world»?

The present work declined such intermediate positions for the sake of clarity. The alternatives remain open for investigation.

Question 2: place of VR-Forms among the foundations

VR-Forms positions itself as a minimalist operational ontology with formal extension. Close positions: Feferman's predicativism (FOM community), Bishop's countable constructivism, Bridgman's operationalism. VR-Forms differs from each, but the precise place on this map requires a separate philosophical study.

Question 3: the next work of the cycle

VR-Forms is the fourth work of the cycle. Open candidates for the fifth:

- (a) **VR-Audit** — systematic analysis of classical mathematics through the lens of VR-Forms;
- (b) **VR-Categories** — categorical reformulation of VR-Sets and VR-Forms;
- (c) **VR-Physics** — operational reformulation of the foundations of physics;
- (d) **VR-Logic** — detailed elaboration of the A1-duality into a self-standing operational logic.

The natural choice is VR-Audit, as a direct application of the VR-Forms apparatus. VR-Categories and VR-Physics presuppose more extended programmes. VR-Logic is a parallel branch developed independently.

VIII.4. Conclusion

Completion of the VR cycle

The VR cycle in its current form — four works:

- (1) **VR. A Formal System** — arithmetic on minimal ontology.
- (2) **VR-Numbers** — numerical extensions over minimal ontology.
- (3) **VR-Sets** — set theory; for the first time reveals the boundary of operational ontology (countability of the universe).
- (4) **VR-Forms** — a formal language for speech about what lies beyond the boundary. Closes the ontological contour: now VR can speak about everything, having in its ontology only $\emptyset$ and the operational.

The slogan of the cycle, carried through the four works:

«Only $\emptyset$ is. Everything else either acts or is said. Action is operational and relates to $\emptyset$; speech is formal and refers to nothing by necessity.»

What the cycle demonstrates

The VR cycle in its completed form demonstrates: **ontological minimalism is possible and workable**. On the sole primitive $\emptyset$ are built arithmetic, numerical extensions, set theory; for the description of the non-operational a formal register is introduced, not violating the minimalism. No additional ontological commitments are accepted; nothing of classical mathematics or speech in general is left without a place.

This is a philosophical position carried through a formal apparatus. It does not claim exclusivity: classical ZFC, constructivism, formalism, categorical set theory — all remain legitimate alternatives. But the VR cycle shows that minimal operational ontology is a **substantive and consistent option**, capable of serving as a foundation for mathematics and as a language for speech in general.

What the cycle does not do

The cycle does not prove the ontological reality of $\emptyset$ — this is an ontological position, not a theorem. The cycle does not deny ontologies other than VR — it only clearly fixes its own and works relative to it. The cycle does not claim instrumental superiority over classical foundations — the programme of VR-Audit is precisely what is intended to systematically check in which cases VR provides something beyond the known.

The mature role of VR is not as an alternative to classical foundations, but as a **transparent and minimalist point of view**, relative to which classical foundations receive a clear classification (through VR-Audit). This is methodological value, not mathematical revolution.

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